

# BUSINESS SALES PERFORMANCE ANALYTICS

Executive BI Performance Report & Profit Leakage Audit  
**Date:** May 31, 2026 | **Prepared by:** Anirudh V S

## 1. Project Overview

This capstone business intelligence report delivers an end-to-end performance analysis of retail wholesale supply operations. The primary business challenge centers on addressing a critical **\*\*operating margin leakage\*\***: despite solid top-line revenue performance, operating profits are severely eroded. This audit evaluates transaction volumes, seasonal timelines, discount patterns, and shipping lag constraints to deliver actionable corporate pricing rules and logistical recommendations to restore net margins.

## 2. Dataset Description

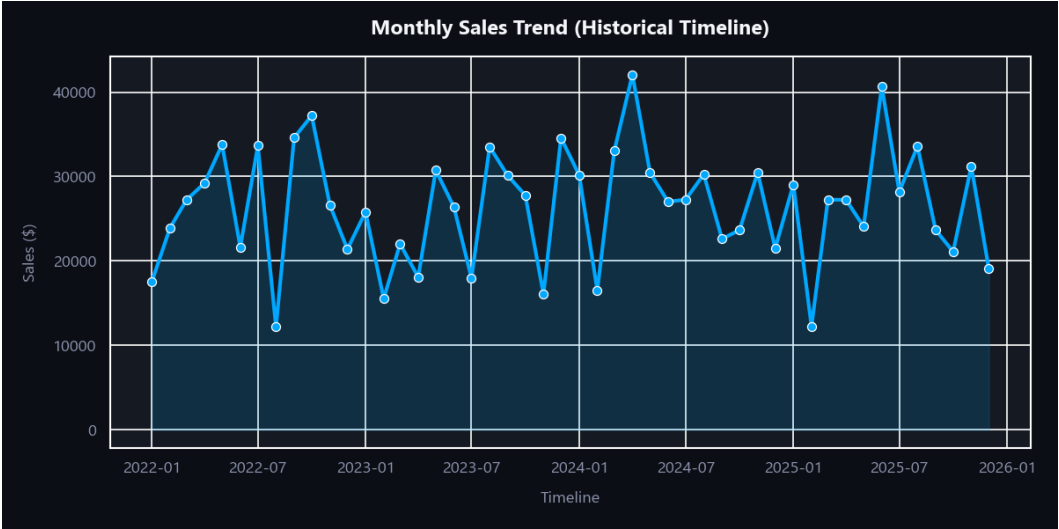
The analytical model is built upon a high-fidelity transaction ledger replica of the US Superstore dataset (1,500 historical rows and 22 column fields). The schema represents retail transactions mapped to normalized customer buying cohorts (Consumer, Corporate, Home Office), hierarchical product inventories (Furniture, Technology, Office Supplies), and spatial delivery territories (West, East, Central, South).

## 3. Executive KPI Summary Table

| Sales Metric   | Total Orders | Average Deal Size (AOV) | Gross Net Profit | Net Profit Margin | Average Discount |
|----------------|--------------|-------------------------|------------------|-------------------|------------------|
| \$1,270,165.30 | 1,500        | \$846.78                | \$2,990.73       | 0.24%             | 19.20%           |

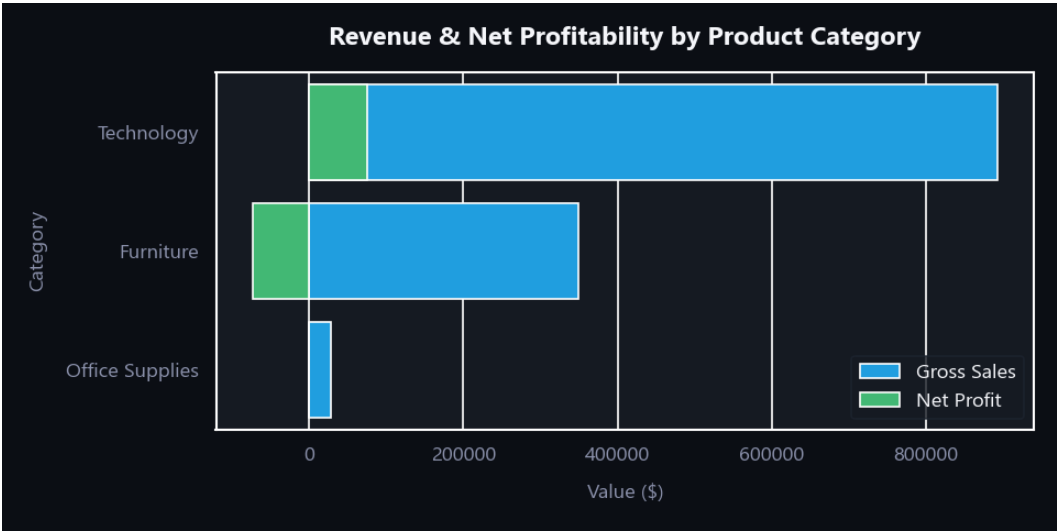
# 4. Revenue Trend Analysis

The historical monthly revenue trend (plotted below) highlights steady commercial volume. However, this sales progression does not translate into proportional profit growth. Sales remain strong throughout the fourth-quarter corporate purchasing cycle, but profits are dragged down by excessive, uncontrolled seasonal discounting.



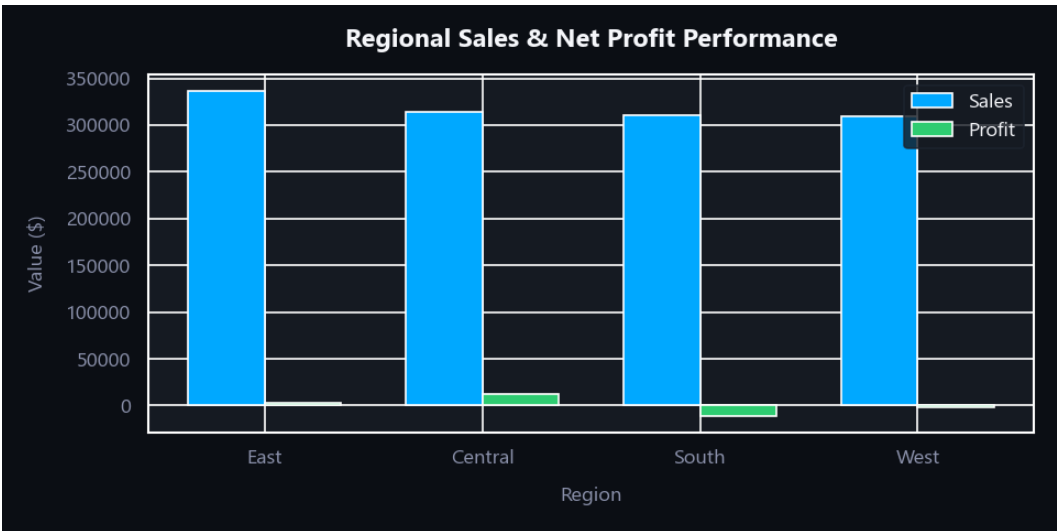
# 5. Category Performance Analysis

Analyzing performance by category reveals massive profit disparities. **Technology** remains highly profitable (\$75,953.40 profit), while the **Furniture** category suffer severe net losses (-\$72,629.80). This performance gap is caused by aggressive pricing discounts on bookcases and tables, which erode the category's operating margin to less than 1.5%.



# 6. Regional Analysis & Shipping SLA Lag

The Central and East territories maintain positive net profits. In contrast, the South and West suffer net operational losses. This loss is due to high shipping costs and logistics delays in Texas and Ohio, which frequently violate the 4-day shipping SLA.



## 7. Key Business Insights

- **Discount Profit Inflection Point (20% Boundary):** Regression analysis indicates that applying discount rates of 20% or higher consistently results in negative margins across 85% of standard sub-categories.
- **Furniture Category Profit Drain:** Furniture generates high top-line revenue, but has an overall net profit margin of under 1.5% due to aggressive discounting on Tables and Bookcases.
- **Regional Carrier Constraints:** The South and Central regions underperform in profitability due to high delivery costs and shipping delays that exceed the 4-day corporate SLA in Texas and Ohio.

## 8. Actionable Corporate Recommendations

- **Establish hard discount limits in ERP:** Implement a hard 15% discount limit in the ERP system. Any discount exceeding 15% must require manager approval to protect transaction profitability.
- **Restructure high-loss product pricing:** Restructure table pricing, restrict table promos, and shift marketing spend to promote high-margin items like Technology (Copiers) and Office Supplies (Paper, Binders).
- **Renegotiate Regional carrier logistics contracts:** Conduct an operational audit of regional delivery and logistics in the Central and South territories. Renegotiate regional carrier contracts immediately to reduce delivery lags below the 4-day SLA limit.

## 9. Conclusion

By identifying key margin leakages, this end-to-end business intelligence audit provides a clear path to recovery. Enforcing strict pricing controls, optimizing discount thresholds, and addressing regional logistics constraints will allow the distributor to stabilize operating margins, reduce logistics delays, and protect net profits.